

The Generalized Lubrication Equation as a subgrid contact-line model: theory, numerics, and validation

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Abstract

A moving contact line carries a viscous stress that diverges at the solid, so the interface shape below the smallest resolved scale of a direct numerical simulation (DNS) cannot be captured on the DNS mesh. This document sets out a subgrid treatment of that region built on the Generalized Lubrication Equation (GLE) of Snoeijer, which stays quantitative for steep interface slopes where classical lubrication theory fails. We record the derivation and sign conventions of the arc-length GLE system, the two solvers that share its right-hand side (adaptive Cash–Karp shooting and a fixed-mesh collocation branch tracer), the calibration used to reproduce the dip-coating bifurcation diagram of Snoeijer & Andreotti, the numerical verification of every component, the alternative slippery-wedge equation of Luo & Gao, and the seam that couples the GLE to a Basilisk two-phase DNS at the grid scale. The runtime solver stack is header-only C99 with no external dependencies; the open-source reference-table generator in `gle-ode/reference-generator/` is an in-repository, development-time numerical authority and is not linked into Basilisk.

1 Introduction

The hydrodynamic model of a moving contact line predicts a shear stress that grows like $1/r$ towards the contact line, so the viscous dissipation carries a non-integrable logarithm and no solution exists within continuum hydrodynamics alone [1]. A microscopic mechanism must cut the singularity off; the simplest and the one used here is a Navier slip length λ , below which the no-slip condition is relaxed. The physical consequence is that the interface bends over a range of scales set by λ at the bottom and by the macroscopic geometry (here the capillary length $\ell_\gamma = \sqrt{\gamma/\rho g}$) at the top — six or more decades of arc length for a millimetric bath and a nanometric slip length.

Resolving all of those decades on a single DNS mesh is infeasible. The strategy adopted here divides the labour by length scale. A subgrid model carries the viscous bending of the interface from the slip length up to the DNS grid size Δ ; a Basilisk two-phase DNS carries it from Δ up to the macroscopic scale. The two are coupled at Δ : each DNS time step the subgrid model hands the DNS an *apparent* contact angle at the grid scale, and takes the DNS interface curvature at that scale as its own outer boundary condition. The subgrid wedge never has to be resolved on the DNS mesh.

The subgrid model is the GLE of Snoeijer [4], a long-wavelength theory that — unlike ordinary lubrication theory — remains quantitative when the interface is steep, and reduces to lubrication theory in the small-slope limit. This document covers, in turn: the derivation and conventions of the GLE in the arc-length form used in the code (Section 2); the two

numerical solvers and the calibration to the dip-coating problem (Section 3); the verification of each component against independent references (Section 4); and the GLE $\leftrightarrow$ DNS coupling seam (Section 5).

2 Theory

2.1 From the wedge expansion to the generalized lubrication equation

Classical lubrication theory expands the Stokes equations in the interface slope and is therefore restricted to $|\nabla h| \ll 1$. Snoeijer’s observation is that at low capillary number the physical requirement is not that the interface be flat, but that its *curvature* vary slowly: surface tension is strong enough to keep the shape close to that of a straight wedge, whatever the opening angle [4]. One therefore expands the Stokes flow about the Huh–Scriven similarity solution for a wedge of finite, slowly varying opening angle $\theta(x)$ [1], treating the variation of θ as the perturbation. The small parameter is $\varepsilon = \text{Ca}^{1/2}$, with $\text{Ca} = \eta U^*/\gamma$, and the expansion is self-consistent as long as

$$h \partial_x \theta \ll 1, \quad (1)$$

i.e. the interface angle turns over a length long compared with the local film thickness. There is no restriction on the size of θ itself.

The result is the generalized lubrication equation (Snoeijer [4], Eq. 14),

$$\partial_x \kappa = 3 \text{Ca} \frac{U/U^*}{h^2} F(\theta) + \frac{1}{\gamma} \partial_x \Phi|_{z=h}, \quad \kappa = \frac{\partial_{xx} h}{(1 + \partial_x h^2)^{3/2}}, \quad (2)$$

with the full curvature κ retained (not its small-slope approximation), the body-force potential Φ , and the mobility factor (Snoeijer [4], Eq. 15)

$$F(\theta) = \frac{2}{3} \frac{\tan \theta \sin^2 \theta}{\theta - \cos \theta \sin \theta}. \quad (3)$$

As $\theta \rightarrow 0$, $F(\theta) \rightarrow 1$ and Eq. (2) collapses onto the ordinary lubrication equation, so the GLE is a slope-dependent correction of the viscous term that is exact at leading order in ε for any slope.

2.2 The arc-length system

Parameterising the interface by its curvature keeps Eq. (2) single-valued through overhangs and folds, which the Cartesian form cannot represent. Let s be the arc length along the liquid–gas interface, measured from the contact line, and take the state vector $y = (h, \theta, \omega, \zeta)$, with h the film thickness normal to the plate, θ the local interface inclination, $\omega = \text{d}\theta/\text{d}s$ the curvature, and ζ the distance travelled along the plate. The GLE becomes the first-order system

$$\frac{\text{d}h}{\text{d}s} = \sin \theta, \quad \frac{\text{d}\theta}{\text{d}s} = \omega, \quad \frac{\text{d}\omega}{\text{d}s} = \frac{3 \text{Ca} M(\theta, \mu_r)}{h(h + c_\lambda \lambda)} + G(\theta), \quad \frac{\text{d}\zeta}{\text{d}s} = \cos \theta. \quad (4)$$

Equation (4) is the Chan *et al.* constant-cutoff form. The denominator $h(h + c_\lambda \lambda)$ is the Navier-slip regularisation of the bare h^2 : it removes the contact-line singularity of the viscous term while recovering h^2 for $h \gg \lambda$. The cutoff coefficient $c_\lambda = c(\theta_e, \mu_r)$ follows from Cox–Voinov matching with slip [8]. The value $c_\lambda = 3$ is the classical one-fluid, small-angle convention, not a universal constant. The implementation accepts an explicit coefficient or resolves it once from (θ_e, μ_r) using the reference workflow of Section 4.1; it remains constant along that case’s trajectory. The direct Luo–Gao model described there uses the same state vector and outer boundary condition but replaces the viscous term rather than introducing c_λ .

2.3 The two-fluid Huh–Scriven mobility

For a passive outer fluid the mobility is $F(\theta)$ of Eq. (3). The code carries the full two-fluid Huh–Scriven wedge factor, which accounts for viscous dissipation in the displaced phase at a gas/liquid viscosity ratio μ_r [7]:

$$M(\theta, \mu_r) = \frac{2 \sin^3 \theta [\mu_r^2 f_1(\theta) + 2\mu_r f_3(\theta) + f_1(\pi - \theta)]}{3 [\mu_r f_1(\theta) f_2(\pi - \theta) + f_1(\pi - \theta) f_2(\theta)]}, \quad (5)$$

with

$$f_1(\theta) = \theta^2 - \sin^2 \theta, \quad f_2(\theta) = \theta - \sin \theta \cos \theta, \quad f_3(\theta) = \theta(\pi - \theta) + \sin^2 \theta. \quad (6)$$

The *plus* sign in the denominator of Eq. (5) is load-bearing: a minus sign there flips the sign of the entire viscous term at small μ_r , and every earlier C port in this repository’s history carried that error. In the one-fluid limit ($\mu_r = 0$) the general form reduces to

$$M(\theta, 0) = \frac{2 \sin^3 \theta}{3 f_2(\theta)} = \cos \theta F(\theta), \quad (7)$$

and $M \rightarrow 1$ as $\theta \rightarrow 0$, recovering the classical result. The mobility is not named $\mathbf{f}$ in the code because in Basilisk $\mathbf{f}$ is the volume-of-fluid volume fraction.

2.4 Sign conventions and the flat-film consistency check

We take $\text{Ca} = \eta U / \gamma > 0$ for a *receding* contact line — the dip-coating configuration in which the plate is withdrawn from the bath — and $\text{Ca} < 0$ for an advancing line. The consistency check is the flat-film limit of the vertical plate: setting $\theta \rightarrow 0$ and $\omega' \rightarrow 0$ in Eq. (4) balances the viscous term against gravity and gives the entrained-film thickness

$$h_{\text{film}} = \sqrt{3 \text{Ca}} \ell_\gamma, \quad (8)$$

the well-known result recovered by Snoeijer [4] (Fig. 3 of that paper) for a film dragged down a vertical plate against a receding contact line.

Gravity enters through

$$G(\theta) = -g^* \cos \theta \quad (\text{vertical plate}), \quad G(\theta) = +g^* \sin \theta \quad (\text{horizontal plate}), \quad (9)$$

with $g^* = (\ell_{\text{unit}} / \ell_\gamma)^2$ the gravity prefactor: $g^* = 1$ when lengths are measured in capillary lengths, and $g^* = 0$ disables gravity (the length unit is then free). Retaining g^* as a general scaling rather than fixing it to unity makes the gravity-rescaling invariance of Section 4 an explicit test rather than an identity. In the vertical geometry s runs from the contact line *down* towards the bath, so the elevation is $z(s) = \Delta - \zeta(s)$ and the static far field obeys the hydrostatic balance $\omega = g^* z$.

2.5 The oscillatory film tail and small negative angles

On the upper (film) branch of the dip-coating problem the interface joins the meniscus through an oscillatory tail, the classical dimple oscillation of a Landau–Levich film, in which θ dips slightly below zero. The mobility must therefore be continued to small negative angles, and how it is continued depends on μ_r . In the one-fluid limit $M(\theta, 0) = 2 \sin^3 \theta / [3 f_2(\theta)]$ is analytically *even* in θ , so the exact even extension $M(\theta) = M(|\theta|)$ applies, with $M(0) = 1$. For $\mu_r > 0$ the mobility is *not* even: near $\theta = 0$ it carries a genuine linear term,

$$M(\theta, \mu_r) = 1 + \frac{3 \mu_r \theta}{2\pi} + \mathcal{O}(\theta^2), \quad (10)$$

so the raw two-fluid formula is evaluated at the *signed* θ (the apparent $0/0$ at $\theta = 0$ is removable and set explicitly to $M = 1$). Because f_1 and f_2 lose all significant digits to cancellation for small arguments ($f_1 \sim \theta^4/3$, $f_2 \sim 2\theta^3/3$), both switch to their Taylor series below $\theta = 0.02$, which is also more accurate than the direct trigonometric form there.

2.6 The dip-coating boundary-value problem

The steady meniscus at a withdrawn plate is a two-point boundary-value problem for the system (4). At the inner boundary, the contact line,

$$h(s_0) = h_0, \quad \theta(s_0) = \theta_e, \quad \omega(s_0) = \omega_0 \text{ (unknown)}, \quad (11)$$

with θ_e the microscopic (equilibrium) contact angle. The integration starts at a small but finite $s_0 \sim \lambda$ rather than at $s = 0$ because the curvature diverges logarithmically at the contact line even with slip; the contribution of the interval $(0, s_0)$ is a bounded logarithm absorbed into the calibration.

The outer boundary condition is that the trajectory has landed on the static meniscus manifold by the matching thickness $h = H$. Static solutions ($\text{Ca} = 0$, vertical plate) conserve the first integral $\frac{1}{2}\omega^2 + g^* \sin \theta$; the branch connected to a flat bath ($\omega \rightarrow 0$, $\theta \rightarrow \pi/2$) therefore satisfies

$$\omega|_{h=H} = \sqrt{2g^*(1 - \sin \theta|_{h=H})}. \quad (12)$$

Matching to Eq. (12) at a moderate H (the code uses $H = 5$) is legitimate because the viscous term in Eq. (4) has by then decayed like $3\text{Ca}M/h$. The meniscus rise follows from the hydrostatic balance $\omega = g^*z$ on the manifold,

$$\Delta = \zeta|_{h=H} + \omega|_{h=H}/g^*, \quad (13)$$

and the apparent contact angle from the classical meniscus-rise relation $\Delta = \sqrt{2(1 - \sin \theta_{\text{app}})/g^*}$ (Snoeijer & Andreotti [11], Eq. 4), i.e.

$$\theta_{\text{app}} = \arcsin\left(1 - \frac{1}{2}g^*\Delta^2\right). \quad (14)$$

Setting $\theta_{\text{app}} = 0$ in Eq. (14) gives the useful reference level $\Delta = \sqrt{2/g^*}$, i.e.

$$z_{\theta_{\text{app}}=0} = \sqrt{2} \ell_\gamma \quad (15)$$

in capillary-length units. For the finite effective slips used here the branch crosses this level before its saddle-node: the legacy fold is $\Delta^* = 1.43908$, corresponding through Eq. (14) to $\theta_{\text{app}} = -2.03^\circ$. The digitised fold is at 1.44617. Above Ca^* no steady meniscus exists and a film is entrained: the Landau–Levich transition, at which the upper branch approaches the entrained-film thickness of Eq. (8). The existence of this maximum speed, and the avoided-critical-behaviour structure around it, were established for this system by Eggers [12] and by Snoeijer *et al.* [5]; the experimental branch is that of Delon *et al.* [6].

3 Numerical implementation

Two complementary solvers share the right-hand side (4). Both are header-only C99 and need only `libm` — no GSL, no SUNDIALS — so the same headers compile under a host compiler and under Basilisk’s `gcc`.

3.1 Adaptive Cash–Karp integration

The trajectory spans the slip scale $\lambda \sim 10^{-6}\ell_\gamma$ to the capillary scale in one integration, and near the contact line $d\omega/ds \sim 1/s$, which a fixed step cannot follow. The integrator is an embedded Cash–Karp Runge–Kutta 5(4) pair with proportional step control. The step error is measured in the mixed norm

$$E = \left[\frac{1}{4} \sum_{i=1}^4 \left(\frac{\text{err}_i}{\text{atol} + \text{rtol} \max(|y_i|, |y_i^5|)} \right)^2 \right]^{1/2}, \quad (16)$$

a step is accepted when $E \leq 1$, and the next step is scaled by $\min(5, \max(0.2, 0.9 E^{-1/5}))$; the same comparison rejects a NaN error norm, which fails every ordinary inequality. Integration terminates on a *thickness event* $h = H$, localised to a relative tolerance of 10^{-13} on h by bisection on the final step size. This is how the outer matching point of the boundary-value problem is hit exactly.

3.2 Single shooting on the contact-line curvature

The static-manifold condition (12) reduces the boundary-value problem to a single scalar equation in the unknown ω_0 ,

$$\mathcal{R}(\omega_0) \equiv \omega|_{h=H} - \sqrt{2g^*(1 - \sin \theta|_{h=H})} = 0, \quad (17)$$

solved by a damped Newton iteration with a central finite-difference derivative, falling back to geometric bracket expansion and bisection on stagnation. Trajectories that leave the physical domain before reaching H return signed penalties chosen so that a collapsing interface ($\theta \rightarrow 0$, $h \rightarrow 0$) reads as ω_0 too small and a fold-over ($\theta \rightarrow \pi$) as ω_0 too large; bisection then still brackets.

The conditioning of Eq. (17) sets a floor on the achievable residual. The flat-bath fixed point of Eq. (4) has a growing mode, so a perturbation at the contact line is amplified downstream like e^s . The matching therefore cannot be pushed to large H without losing all precision, and even at $H = 5$ the residual floor with the default integrator tolerances is $\sim 10^{-8}$; the Newton tolerance is set at 5×10^{-8} accordingly. (The alternative far-field condition $\omega(s_{\max}) = 0$, used only for the cross-validation of Section 4, has $\mathcal{O}(1)$ sensitivity in ω and supports a tolerance of 10^{-11} .) Shooting is the fast single-solve path and the engine of the DNS coupling; on the lower branch it is reliable up to roughly 0.6 Ca^* , beyond which the residual window in ω_0 narrows exponentially towards the fold.

3.3 Collocation branch tracing through the fold

Shooting cannot trace the full bifurcation diagram. In the (ω_0, Ca) plane the branch makes a hairpin at the fold that is narrower than the shooting map can resolve: the forward flow *compresses* the separation between neighbouring branches like e^{-s} while amplifying noise like e^{+s} . Deep on the upper branch the trajectory carries a film section of length $\sim \Delta$ whose exponential dichotomy makes shooting ill-conditioned in *both* directions. The chart (ω_0, Ca) is therefore degenerate at the fold.

The branch tracer instead discretises the whole problem on a *fixed* mesh and parameterises solutions by the meniscus rise Δ , which is monotone along the entire branch. Nodes $s_i = s_0 (s_{\text{end}}/s_0)^{\tau_i}$ use a hybrid grading, logarithmic below $s = 0.3$ (to resolve the slip-to-mesoscale decades near the contact line) and uniform above (to resolve the film and meniscus foot, whose widths are $\mathcal{O}(\sqrt{\text{Ca}})$); only the uniform part dilates with the unknown domain length s_{end} . The cell residuals are the second-order implicit midpoint rule,

$$y_{i+1} - y_i - (s_{i+1} - s_i) \mathbf{f}\left(\frac{1}{2}(y_i + y_{i+1})\right) = 0, \quad (18)$$

closed by the contact-line conditions, the static-manifold condition (12) at node N , the definition $h_N = H$ (which fixes s_{end}), and the target rise $\zeta_N + \omega_N/g^* = \Delta^*$. The unknowns are the $4(N+1)$ node states together with Ca and s_{end} ; the equations are the $4N$ collocation residuals plus six conditions. The system is square.

The Jacobian is block-bidiagonal of bandwidth seven with two dense border columns ($\partial/\partial \text{Ca}$ and $\partial/\partial s_{\text{end}}$, formed by finite differences of the full residual) and two border rows (the h_N and Δ conditions). It is solved by the bordering algorithm: banded LU with partial pivoting on the square band part, three back-solves, and a 2×2 reduced system for the two border unknowns.

Because Δ is monotone along the branch, marching Δ^* with a secant predictor and an adaptive step (halved on a failed corrector, grown by $1.4\times$ on fast convergence) walks straight through the fold with no turning-point handling at all. This is exactly what the mesh re-adaptation of SciPy’s `solve_bvp` inside a continuation loop could not do — the failure mode of the historical Python continuation (issues #14 and #16 of the repository) — because a mesh that moves between Newton iterations makes the extended continuation system inconsistent. A fixed mesh keeps it consistent. The construction is a stabilised multiple-shooting march in the sense of local parameterisation continuation [17].

3.4 Parameters and calibration for Fig. 4b

The Fig. 4b reproduction has two fitted microscopic inputs, the equilibrium angle θ_e and an effective slip parameter λ , with the legacy cutoff convention fixed explicitly at $c_\lambda = 3$. They are calibrated against the vector-digitised theory curve of Fig. 4b of Snoeijer & Andreotti [11], not measured independently:

- $\theta_e = 53.46^\circ$, fixed by the static foot $\Delta(\text{Ca} \rightarrow 0) = \sqrt{2(1 - \sin \theta_e)} = 0.627$ at the contact line. The implemented boundary starts at finite s_0 and adds $s_0 \cos \theta_e$, a 4.44×10^{-6} offset for the default case;
- $\lambda/\ell_\gamma = 7.46 \times 10^{-6}$, fixed by the digitised fold $\text{Ca}^* = 10.54 \times 10^{-3}$. The prediction is only logarithmically sensitive to λ , so the fold calibration absorbs the chosen $c_\lambda = 3$ cutoff convention into this fitted value. It must not be read as an independently measured physical Navier length.

Because the microscopic cutoff is calibrated to Ca^* , recovering Ca^* from the traced branch is circular by construction and is not a test. The genuine out-of-sample checks are the fold height $\Delta^* = 1.440$ (digitised 1.446, not fit to) and the full branch shape, including the upper-branch approach to the Landau–Levich asymptote, which carries no further free parameters.

The one-shot reproduction additionally enforces an external branch-shape gate. Resampling the legacy branch and independently digitised theory curve onto 421 uniformly spaced heights gives maximum $|\Delta\text{Ca}| = 2.012 \times 10^{-4}$ and RMS 1.044×10^{-4} , below fixed budgets of 2.2×10^{-4} and 1.1×10^{-4} . Every generated branch must also reach its requested upper-branch height, so a truncated continuation cannot pass this gate.

The legacy $c_\lambda = 3$ reproduction is retained as provenance. Its effective slip was calibrated to the digitised fold at the reported resolution. For the case-dependent Chan closure and the direct Luo–Gao equation, the new effective slips are fitted to the computed legacy fold $\text{Ca}^* = 1.0544249 \times 10^{-2}$. These are calibrations, not independent validations. Comparisons between the two equations therefore also hold λ , θ_e and μ_r fixed; otherwise a refitted microscopic length can conceal the full formulation difference.

4 Validation

The checks below distinguish numerical verification from physical-model validation. Cross-solver and cross-language tests establish that the two C solvers integrate their stated equations consistently, while mesh convergence and gravity invariance test the discretisation and units. The finite- μ_r Chan closure is checked against the independent inner-Stokes solves at the Q level. The integrated Fig. 4b branch comparison is one-phase; no independent finite- μ_r full-profile benchmark is claimed here.

4.1 Microscopic closure and GLE-model protocol

Numerical authority. At $\mu_r = 0$, a dedicated public generator solves Scott’s singular integral equation for the wall-stress function and evaluates $Q_i = k_\infty^{-1} \int \Gamma \, d\rho$. Subtracting the value at the

collocation point regularises the logarithmic kernel, while Scott’s analytic K_0 series supplies the truncated-domain tails. This produces the separate one-phase table and convergence evidence described below.

For finite μ_r , the two-phase reference generator solves the Stokes equations in a straight wedge with the same constant Navier slip length on both solid sectors. The outer boundary is the Huh–Scriven wedge velocity and the inner solution is regularised by slip. From the computed wall force it fits, with μ_1 the reference-phase viscosity,

$$\frac{W}{\mu_1 U} = \frac{3M(\theta_e, \mu_r)}{\sin \theta_e} \ln \frac{R}{\lambda} + H, \quad Q = 1 + \frac{\sin \theta_e}{3M(\theta_e, \mu_r)} H, \quad (19)$$

and converts Q to the Chan cutoff through

$$c_\lambda = \sin \theta_e \exp(1 - Q). \quad (20)$$

The logarithmic slope is checked against its analytic coefficient, while mesh, inner and outer radii, and the logarithmic fitting window are refined independently. Only converged values are frozen on the $(\theta_e, \log \mu_r)$ table. Thus the generator is the numerical authority for the chosen equal-constant-slip microscopic model; it is not a claim that c_λ is universal across other microscopic boundary conditions.

Dependency-free runtime closure. The production C code does not solve the Stokes problem. It consumes the frozen table and resolves c_λ once after the case parameters are known. The automatic policy is ordered deliberately: (i) the Scott–Hocking $\mu_r = 0$ reference branch, obtained from Hocking’s integral problem as evaluated by Scott [9, 10]; (ii) the corrected right-angle branch; (iii) interpolation in Q over the convergence-checked table; and (iv) the explicit Luo–Gao value of Q as an *approximate* fallback with that provenance exposed to the caller. Interpolating Q , rather than c_λ , preserves the additive Cox matching constant. The fallback is used only outside available reference data; it is not promoted to numerical authority. The Scott–Hocking implementation consumes 70 direct solutions of Scott’s singular integral problem and the analytic Hocking right-angle anchor. Each node is compared with coarser and wider-domain solves; the largest recorded sensitivity is 6.618×10^{-5} in Q_i . Direct solutions at all interior cell midpoints and in both asymptotic endpoint intervals give 71 independent checks, with maximum interpolation discrepancy 4.240×10^{-5} and maximum discrepancy-plus-solve-sensitivity budget 1.793×10^{-4} . Scott’s published table is reproduced to 3.106×10^{-6} . The arbitrary-angle result is therefore a numerically converged reference, not an analytic exact formula; only the right-angle anchor is analytic exact.

The frozen half-plane contains 270 converged values on a refined 27×10 grid over $30^\circ \leq \theta_e \leq 150^\circ$ and $10^{-2} \leq \mu_r \leq 1$; phase exchange extends the runtime range to $10^{-2} \leq \mu_r \leq 10^2$ without extrapolation. Runtime interpolation uses a four-node cubic polynomial in θ_e and a five-node quartic polynomial in $\log_{10} \mu_r$. A separate FEM solve at the centre of every one of the 234 table cells gives $\max |Q_{\text{interp}} - Q_{\text{FEM}}| = 3.8919 \times 10^{-4}$. Adding the recorded convergence sensitivities of the checkpoint and the 20 interpolation nodes with absolute Lagrange weights gives a worst propagated budget of 1.8310×10^{-3} . The largest table-node sensitivity is 5.6403×10^{-4} ; the right-angle discrepancy is 1.4458×10^{-4} . These are empirical cell-centre and refinement checks, not a global interpolation-error bound inside every cell. The raw $\mu_r = 1$ column obeys phase exchange to 8.98×10^{-7} ; reflected pairs are averaged only in the emitted C header to impose the exact identity, while the unmodified FEM values remain public in the CSV.

Corrected right-angle branch. The correction identified by Luo & Gao [16] restores the phase-exchange symmetry $c(\pi/2, \mu_r) = c(\pi/2, 1/\mu_r)$. Representative values are

μ_r	0	0.02	1	10
$c(\pi/2, \mu_r)$	1.1229189671	1.20925446	2.05206963	1.47616934

The limits $\mu_r \rightarrow 0$ and $\mu_r \rightarrow \infty$ both give $\exp(\ln 2 - \gamma_E) = 1.1229189671$. The finite- μ_r entries are regression anchors evaluated using the published four-significant-figure constant $h_b = -1.539$; their final displayed digits do not imply greater physical precision.

Two GLEs, not two spellings of the same closure. The Chan equation uses the case-constant scalar from Eq. (20). The direct Luo–Gao equation instead applies the local slippery-wedge approximation of Luo and Gao and evaluates [16, Eq. 3.16]

$$\left. \frac{d\omega}{ds} \right|_{\text{visc}} = \frac{\text{Ca}}{h} \frac{Dh \sin^2 \theta + E\lambda \sin^3 \theta}{Ah^2 + B\lambda h \sin \theta + C\lambda^2 \sin^2 \theta}, \quad (21)$$

where, using Eq. (6) [16, Eq. 3.17],

$$\begin{aligned} A &= \mu_r f_1(\theta) f_2(\pi - \theta) + f_1(\pi - \theta) f_2(\theta), \\ B &= 2 \sin^2 \theta [\mu_r f_1(\theta) + f_1(\pi - \theta)] + 2(\mu_r + 1) f_2(\theta) f_2(\pi - \theta), \\ C &= 4 \sin^2 \theta [\mu_r f_2(\theta) + f_2(\pi - \theta)], \\ D &= 2 \sin \theta [\mu_r^2 f_1(\theta) + 2\mu_r f_3(\theta) + f_1(\pi - \theta)], \\ E &= 4 \sin \theta [\mu_r^2 f_2(\theta) + \mu_r \pi + f_2(\pi - \theta)]. \end{aligned} \quad (22)$$

All five coefficients use the *local* angle $\theta(s)$. The equation contains no c_λ . Both equations reduce to the same Huh–Scriven outer mobility for $h \gg \lambda$, but differ within the slip-to-mesoscale region. In the one-phase limit Luo–Gao matching gives $c_{\text{LG}} = 2 \sin^3 \theta_e / f_2(\theta_e)$; inserting that number into the Chan equation matches one asymptotic constant but does not reproduce Eq. (21), which updates the factor as $\theta(s)$ evolves. Chan with a Scott–Hocking, right-angle, or numerical-table value of Q is the reference-matched route when one of those sources covers the case. Direct Luo–Gao is the fully explicit local alternative when no table value is available or that local closure is the object of study; its slippery-wedge construction remains an approximation. Their difference is reported as formulation uncertainty. The published two-phase expression has the wedge domain $0 < \theta < \pi$; only its $\mu_r = 0$ reduction has an exact even extension through the small negative angles in an oscillatory film tail. The direct-model Fig. 4b comparison is consequently made in that one-phase limit. At finite μ_r , a trajectory that reaches $\theta \leq 0$ must use the Chan route rather than extrapolate Eq. (21).

The two routes are measurably different even before solving an interface profile. At the 270 frozen finite- μ_r nodes, the explicit Luo–Gao matching approximation gives $Q_{\text{LG}} - Q_{\text{FEM}}$ between -0.95525 and -0.00710 , with RMS 0.22665. For example,

θ_e at $\mu_r = 0.1$	60°	90°	120°
Q_{FEM}	0.121237	0.610636	0.689767
Q_{LG}	0.086959	0.522681	0.525533
$c_{\text{LG}}/c_{\text{FEM}}$	1.03487	1.09194	1.17849

Figure 1 shows the full nodewise comparison. Each square is one converged FEM solve on the deliberately nonuniform grid; no interpolation or contour reconstruction is used in this figure.

Thus Chan with the numerical table is preferred inside the certified domain when equal constant slip is the intended microscopic model. Direct Luo–Gao is preferable when its local slippery-wedge closure is itself desired or a dependency-free explicit model outside the frozen domain is more valuable than reference matching; it should not be described as the more accurate inner solution by default. The comparison is specific to this microscopic model and table domain, not a universal ranking of contact-line regularisations.

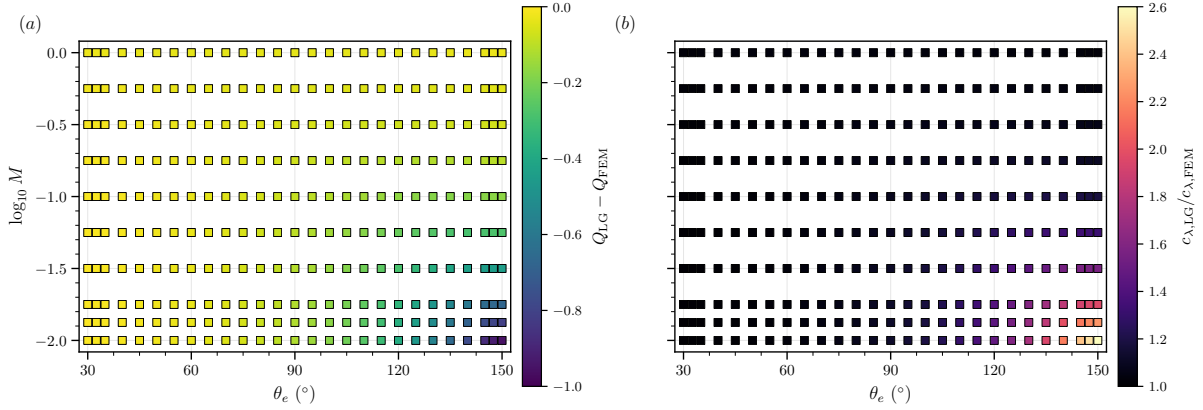


Figure 1: Finite-viscosity matching comparison over the 270 certified inner-Stokes table nodes. (a) Difference between the explicit Luo–Gao approximation and the FEM reference matching constant. (b) Corresponding ratio of Chan cutoff coefficients, using $c_\lambda = \sin \theta_e \exp(1 - Q)$. The ranges are $-0.95525 \leq Q_{LG} - Q_{FEM} \leq -0.00710$ and $1.00713 \leq c_{\lambda, LG}/c_{\lambda, FEM} \leq 2.59932$; the largest mismatch is at $\theta_e = 150^\circ$, $\mu_r = 10^{-2}$. Markers are the actual FEM nodes and no values are inferred between them.

Calibration-honest Fig. 4b comparison. Two comparisons answer distinct questions. First, the legacy effective λ is fitted to the digitised fold and each new model is fitted to the computed legacy fold. This tests only the out-of-sample fold height and branch shape; fold agreement itself is circular. Second, running Chan with reference-derived c_λ and direct Luo–Gao at identical input (λ, θ_e, μ_r) exposes differences that the independent fits would hide. This is the total formulation difference, including the models’ distinct matching constants and local viscous terms; agreement of effective matching constants alone would still not guarantee identical Fig. 4b branches. At the implemented $s_0/\lambda = 1$, the common-input fold shift is 0.41690%. Matched starts at $s_0/\lambda = 0.1, 0.01$ and 0.001 give 0.6289%, 0.6819% and 0.6903%, respectively. Thus the sub-percent conclusion is stable, while the single 0.41690% value is specific to the finite-start convention; no common finite- s_0 inner correction is included.

Mobility. The two-fluid mobility (5) agrees with the Python reference to machine precision across $\theta \in (0, \pi)$ and $\mu_r \in [0, 1]$. Below $\theta = 0.02$ the C series form is in fact *more* accurate than the direct trigonometric formula, which loses precision to the cancellation in f_1 and f_2 .

Slip cutoff. The right-angle matching helper reproduces the analytic one-fluid value

$$c(\pi/2, 0) = \exp(\ln 2 - \gamma_E) = 1.1229189671.$$

The corrected finite-viscosity-ratio values and phase-exchange symmetry are listed above. At general angle, every runtime result carries the name and status of the reference branch, table interpolation, or approximate fallback which supplied it.

Interface profile. Against the historical SciPy `solve_bvp` reference solving the identical boundary-value problem, the interface angle $\theta(s)$ agrees to a relative error below the enforced 2.3×10^{-4} limit throughout the well-conditioned window. The automated gate also requires the thickness error below 5×10^{-4} and the apparent-angle difference below 10^{-3} degrees. The far field is a genuinely soft condition: the gain $d\theta(10^6)/d\omega_0 \sim 5 \times 10^7$, so the two solutions may drift apart in θ at very large s even though the two right-hand sides agree to machine precision. This is a property of the boundary-value problem, not of either implementation.

Shooting versus collocation. At equal Ca the two solvers agree to $\sim 2 \times 10^{-6}$, confirming that the fixed-mesh march and the adaptive shot converge to the same solution branch.

Mesh convergence of the fold. Refining the collocation mesh at $N = 1250, 2500, 5000$ cells, with an initial continuation step $\Delta\Delta = 5 \times 10^{-4}$ capped at 2×10^{-3} , gives the fold location

N	Ca^*
1250	1.0544514×10^{-2}
2500	1.0544248×10^{-2}
5000	1.0544181×10^{-2}

The successive differences fall by a factor of about four as the mesh spacing halves, confirming the second-order convergence of the implicit-midpoint discretisation.

Linear algebra. The bordered banded-LU solve agrees with a dense reference factorisation to $\sim 10^{-14}$, and the Cash–Karp tableau reproduces its published coefficients exactly.

Gravity-rescaling invariance. Rescaling the gravity prefactor to $g^* = 4$ should halve both the slip length and the meniscus rise (in the new length unit) while leaving Ca invariant, since Ca is dimensionless. The traced branch reproduces this to 4.5×10^{-6} , a direct test that the general g^* scaling is carried consistently through both solvers.

Static limit. The $Ca \rightarrow 0$ solve reproduces the static invariant for its implemented finite- s_0 boundary. The computed and contact-line values are

$$\Delta_{s_0} = 0.626995031426, \quad \Delta_{cl} = 0.626990589862.$$

Their 4.44×10^{-6} difference is the geometric offset $s_0 \cos \theta_e$. A regression checks the shifted value, so the small offset is explicit rather than hidden by an “exact” claim.

Figure 2 shows the independently calibrated branches overlaid on the digitised reference, Fig. 3 holds the microscopic inputs fixed between the two equations, and Fig. 4 gives the C-versus-Python cross-validation.

5 The DNS and subgrid GLE coupling

The GLE resolves the contact-line region from the slip length up to the DNS grid size Δ ; the Basilisk two-phase DNS resolves everything from Δ up to the macroscopic scale. The coupling is a per-time-step event that exchanges one scalar in each direction at the grid scale. Each step (`event (i++)`):

1. the DNS interface position and curvature κ_{DNS} are sampled in the finest interfacial cell nearest the plate, including an interface aligned exactly with cell faces;
2. the subgrid GLE is solved on $s \in [s_0, s_\Delta]$ with $h(s_\Delta) = \Delta$, the inner contact-line conditions, and the outer condition $\omega(s_\Delta) = -\kappa_{DNS}$, by shooting on ω_0 . The minus sign follows from $h = y$, $\zeta = -x$ and the phase orientation used by Basilisk’s `curvature()` in this case;
3. the resulting apparent angle $\theta_{app} = \theta(s_\Delta)$ is imposed on the DNS through the height-function `contact_angle()` boundary condition. The height vector is associated explicitly with the VOF tracer, so Basilisk actually applies the dynamic angle.

This is the multiscale strategy of the review [11] applied as a grid-scale boundary condition, in the spirit of the mesh-dependent dynamic contact angle of Afkhami, Zaleski & Bussmann [14]: the imposed angle is a property of the resolved scale, not a fixed material constant.

The seam is implemented and exercised by a bounded Basilisk runtime regression. It starts from the exactly grid-aligned interface that previously evaded the strict volume-fraction filter, confirms refinement to the intended local grid size, and checks that the GLE updates the associated contact angle. During a normal run the capillary number is formed from the plate speed relative to the measured contact-line speed, the actual local cell size is passed to the GLE, and the dynamic angle is stored in a dumped scalar for restart. A grid-convergence study of the coupled curvature and contact-line-position samples remains the problem-specific production qualification. Both the reference-matched Chan model and the direct Luo–Gao model [16] use this same coupling seam. Their equal-physical-slip comparison is kept separate from any per-model calibration. The linearised GLE has separately been checked against the reference data of Kansal *et al.* [15].

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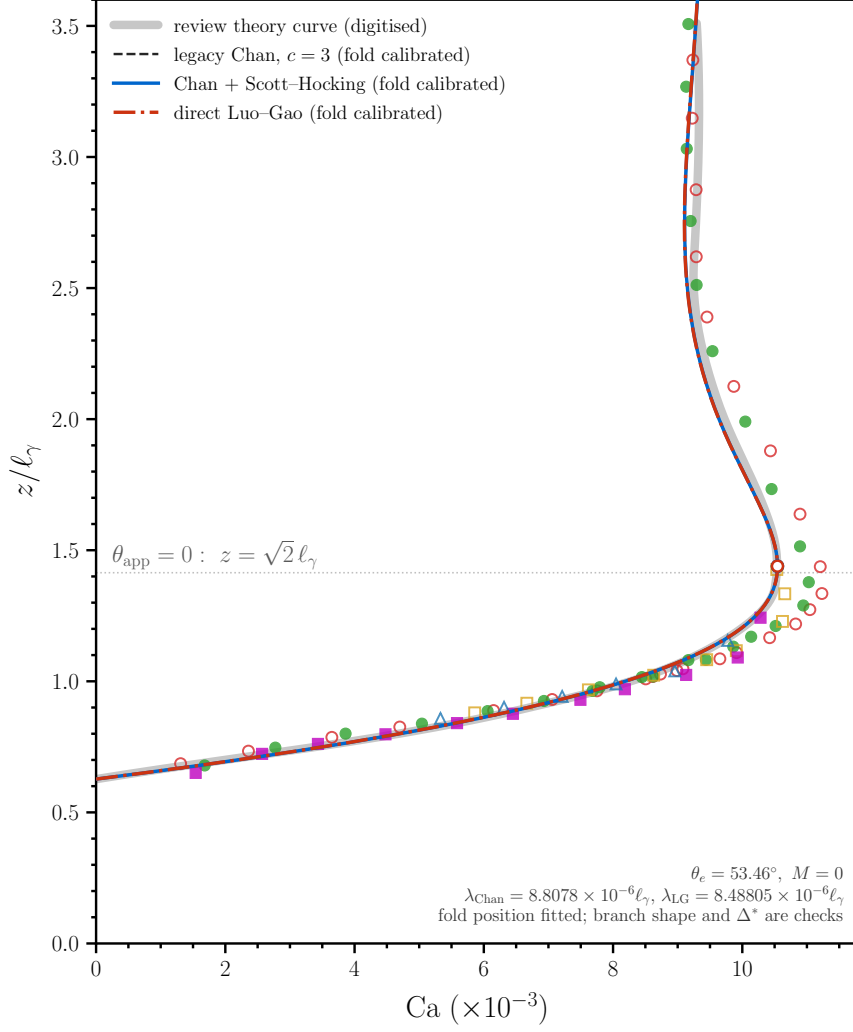


Figure 2: Steady meniscus rise at a receding contact line. Normalised contact-line elevation z/ℓ_γ against capillary number $Ca = \eta U_p/\gamma$ for a partially wetting plate withdrawn from a silicone-oil bath. Dashed black retains the legacy Chan branch with $c_\lambda = 3$, blue uses the Scott–Hocking one-phase closure in the Chan equation, and red uses the direct Luo–Gao equation. Each effective slip length is independently fitted to the same fold; the coincidence there is therefore calibration, while the fold height and branch shape remain comparisons. The thick grey line is the multiscale-lubrication prediction digitised from Fig. 4b of Snoeijer & Andreotti (2013) [11]. Coloured symbols are the experimental realizations of Delon et al. (2008) [6]: each colour is a distinct run, and the markers denote steady menisci measured at fixed plate speed. In the original review figure these symbols are shadowed by thin continuous coloured traces; those are neither theory nor guides to the eye, but the continuously recorded transient elevation $z_{cl}(t)$ of each run mapped into this plane through the instantaneous relative capillary number $\widehat{Ca}(t) = \eta(U_p - dz_{cl}/dt)/\gamma$. Such transients relax adiabatically through the succession of quasi-steady states and therefore retrace the stationary branch; their visible wiggle is measurement noise amplified by the dz_{cl}/dt term. Only the discrete steady-state symbols are digitised here. The dotted $z = \sqrt{2}\ell_\gamma$ line marks the zero-apparent-angle level. At the fitted finite slips the branches cross it before folding near $z/\ell_\gamma = 1.439$; above Ca^* no steady meniscus exists and a film is entrained.

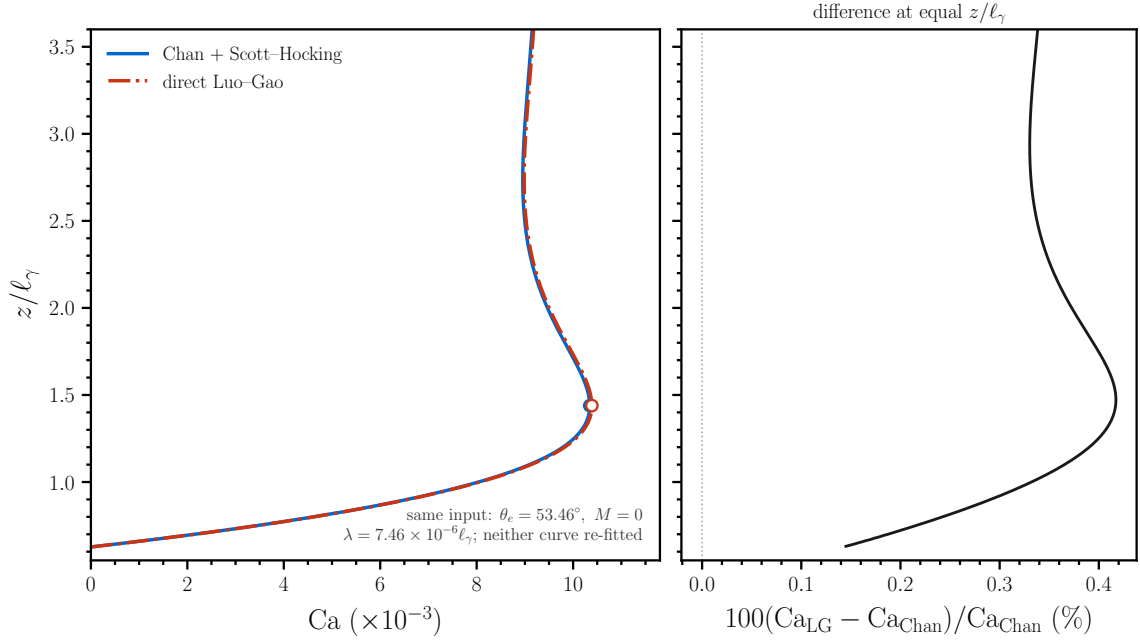


Figure 3: Full-formulation comparison at identical microscopic inputs: $\theta_e = 53.46^\circ$, $\mu_r = 0$, and $\lambda/\ell_\gamma = 7.46 \times 10^{-6}$, with neither equation re-fitted. Left: the Chan equation with the Scott-Hocking closure and the direct Luo-Gao equation. Right: their relative capillary-number difference at equal meniscus rise. The direct Luo-Gao fold is 0.41690% above the Chan fold for this common input and the implemented $s_0 = \lambda$ start. Matched s_0/λ refinement approaches a 0.690% shift, so the robust result is sub-percent rather than the cutoff-specific headline value. The difference includes both the matching constants and the local viscous terms.

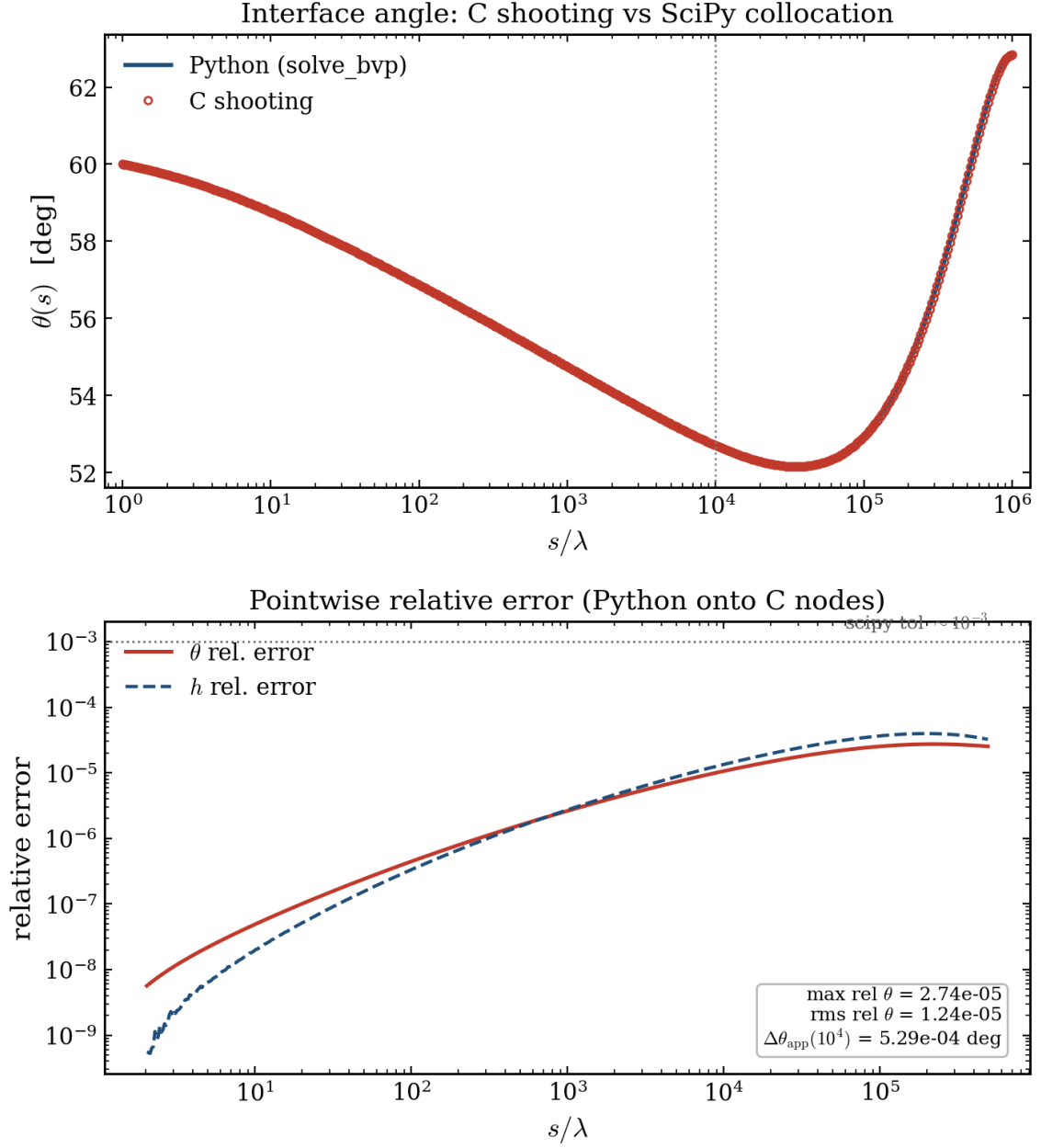


Figure 4: Cross-validation of the C solver against the historical Python `solve_bvp` reference on the identical far-field boundary-value problem. Left: the interface angle $\theta(s)$ on a logarithmic arc-length axis. Right: the relative error between the two solutions on log-log axes. Agreement is excellent where the problem is well conditioned ($s \lesssim 10^4$) and degrades monotonically towards the arc-length cap, a consequence of the soft far-field angle condition rather than of either implementation.